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ORIGINAL ARTICLE

Mathematical modeling and simulated analysis of metal flow behavior during the FGC of ESP rolling process

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Abstract

An accurate description of the metal flow behavior during wedge zone deformation is key to the stability of the endless strip production rolling process. To predict the flatness distribution based on the law of volume invariance, a flatness calculation model of the wedge zone during the flying gauge change (FGC) process was established by considering the metal transverse flow. Furthermore, the effect of different rolling parameters on the flow of metal in the wedge zone was investigated using a combination of this model with a 3D finite element model. In addition, the influences of the reduction rate, roll crown, and friction coefficient on the flatness and metal lateral flow were analyzed and discussed. Finally, the numerical results provided by the finite element model were compared with the computed results of the flatness distribution to illustrate the effectiveness and stability of the proposed model.

Keywords ESP · FGC · Flatness distribution · Metal transverse flow · Finite element model

1 Introduction

Endless strip production (ESP) can provide ultra-thin hot-rolled strips with high quality. It has received increasing attention in recent years [1, 2]. The thickness of the final products is less than 0.8 mm. These can also be substituted for certain cold-rolled products (see Fig. 1). In contrast to the conventional hot-rolling line, the HRM and FM sections are linked by a continuous intermediate slab, and the continuous final coil is split by the flying shear arranged before the downcoiler. Using the flying gauge change (FGC) technology, different ranges of strip thickness can be obtained by the real-time adjustment of the roll gap and velocity [3, 4].

FGC is a technique for modifying the product thickness specifications online during the rolling process. According to the production plan, the roll gaps, roll speeds, tensions, and other factors are adjusted to achieve flexible and continuous modifications in the product thickness specifications. A schematic of the transition specifications is shown in Fig. 2. The wedge zone created by FGC is located between two adjacent product specifications. During the transition from one product specification to another, the zone is rolled into a wedge zone with a thin head and thick tail or a thin tail and thick head owing to the variation in the roll gap. This wedge zone stabilizes the FGC process. Its head and tail indicate the beginning and end, respectively, of the FGC process. The head and tail are called the FGC start and end points, respectively. When the rolling mill captures the FGC start point (wedge zone head) and reaches a predetermined position, the FGC process begins and continues until the FGC endpoint (wedge zone tail) reaches the preset position. The set FGC process ends at this point.

Flatness is an important index for evaluating the quality of strip steel. The flatness of a product is the result of the joint action of various processes. To obtain better flatness, researchers have begun to study the effects of various

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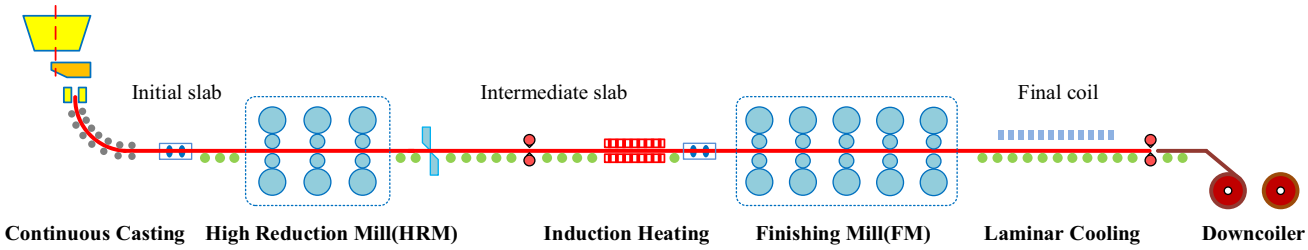


Fig. 1 Layout of an ESP production line

production processes on flatness. These include the impact of pre-straightening on the strip's flatness and residual stress [5], the effect of work roll wear and lateral movement on the quality of the shape of the steel strip [6, 7], and the interference of inter-rack residual stress [8] and strip conveying [9] on the flatness of finished products. In addition, extensive studies have been conducted on the influence mechanism of the rolling parameters on flatness [10–12]. This is effective for controlling the rolling parameters to obtain strip steel with the desired flatness. The FGC process includes variations in the rolling speed, roll-down, and tension. Combined with the high-temperature and high-speed production conditions, these hinder the control of the flatness of the wedge zone. However, the problems that occur in this area affect the quality of the subsequent products and determine whether the production process can proceed smoothly. Therefore, studying the metal flow mechanism in the wedge zone of the FGC process is conducive to reducing the occurrence of production accidents, obtaining products with improved flatness, and improving the stability of the production process.

In early studies, the rolling deformation was generally simplified as a two-dimensional deformation. Moreover, the metal transverse flow of the strip was omitted to simplify the calculation model for the rolling process [13]. However, the metal lateral flow should be considered because it can affect the plastic deformation of the strip [14, 15]. With enhancements in product accuracy requirements, an increasing number of researchers have begun to pay attention to the metal transverse flow of strips and consider the three-dimensional deformation during strip rolling. Allwood [16] established a three-dimensional steady-state model to analyze the strip thickness, metal flow, and stress distribution during hot strip rolling. Li et al. [17] proposed a new method for analyzing metal lateral flow and studied the effects of the bending force, roll shape, and strip crown on the metal lateral flow using FEM. They established a lateral displacement calculation model by interpolation. Moazeni et al. [18] analyzed the metal flow, longitudinal stress, and strain in wide strips using a 3D finite element model. Dixon et al. [19] simplified the equilibrium equations and assumed that the ratio of the transverse and longitudinal strains is constant. Thereby, they obtained an analytical solution for the metal transverse flow of a strip. Ren et al. [20] studied the influence of the metal lateral displacement on the strip shape using the grid method and discussed the effects of reduction and tension on the metal lateral flow during cold thin strip rolling. Chai et al. [21] established a 3D finite element model to obtain a metal transverse flow curve. Furthermore, a flatness calculation model was established while accounting for the metal lateral flow during the rolling process. Yao et al. [22] presented a real-time metal transverse flow model that improved the accuracy of calculating the strip tension

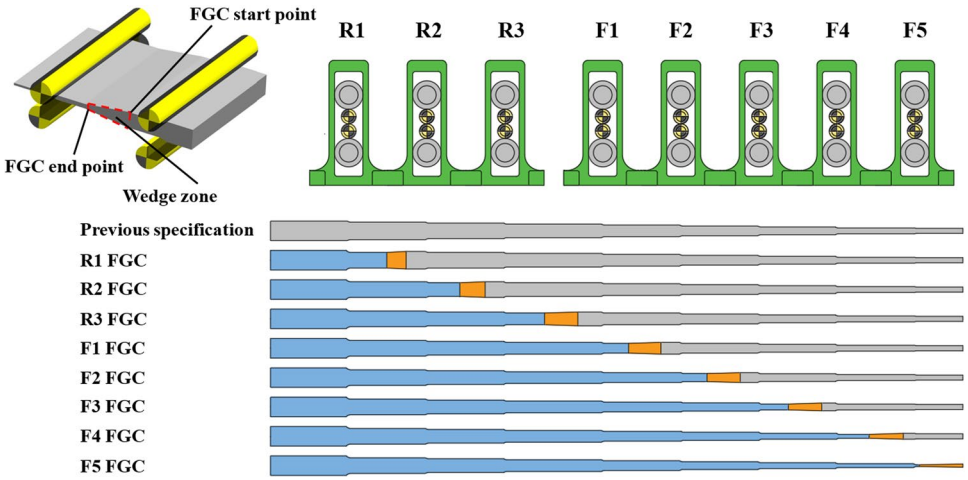


Fig. 2 Schematic diagram of transition specification

and rolling force distribution during the rolling process using the quasi-three-dimensional difference method. They verified the model using the finite element method. Zhao et al. [23] introduced the influence of metal transverse flow on the residual stress. They proposed that metal transverse flow can reduce the risk of buckling deformation of the strip during the strip rolling process. Wang et al. [24] developed a flatness defect calculation model while accounting for metal lateral flow during the rolling process. In addition, they discussed the influence of different reduction rates, thicknesses, and crown ratios (crown ratio is the ratio of the strip crown to the strip thickness) on the metal transverse flow and flatness distribution. However, the aforementioned researchers studied only the metal transverse flow during the normal rolling process and not that in the FGC process.

Few studies have been conducted on the metal flow law in the strip wedge zone deformation process. However, the FGC process is a variable-thickness rolling process, and more research has been conducted on the metal flow law in variable-thickness rolling processes. This can provide a reference for further studies on the FGC process. Kopp et al. [25] proposed the concept of 3D profiled blank through TRB experiments with longitudinal and transverse transitions in the rolling process. Wang et al. [26] proposed a double power function for the transition curves to prevent abrupt variations in the mechanical parameters of tailor-rolled blanks during the rolling process. Zhang et al. [27] studied the metal flow law in the deformation zone of the longitudinal variable-thickness rolling process. Furthermore, they analyzed the influence of the reduction, slope of the transition zone, rolling speed, and other factors. Shafiei et al. [28] introduced the influence of different deformation conditions on the rolling force. They also analyzed the effects of the reduction rate, roll diameter, and friction coefficient on the rolling force during a variable-thickness rolling process. Although the FGC process is similar to the variable thickness rolling process, it is more complex than the standard variable thickness rolling process. Unlike variable-thickness rolling, the roll should be accelerated or decelerated, and the length of the wedge zone should be smaller than the distance between the two stands to ensure rolling stability during the FGC process. Therefore, the metal flow law in the FGC process should be studied further.

The above studies discussed the effects of the metal transverse flow on the rolling force, crown, and flatness during ordinary rolling, and a flatness calculation model considering the metal transverse flow was established. The speed, rolling force, and transition form of the variable-thickness rolling process were analyzed. However, in the FGC process on the ESP line, the flow mechanism of the metal in the wedge transition process at high temperatures is ambiguous. Furthermore, different section profiles, press down, temperatures, rolling speeds, etc., affect the FGC. This results in different flow states of this part of the metal. The existing conventional models are insufficient to effectively describe the flow pattern. In addition to the wide range of influencing factors, methods for predicting the deformation of wedge zones in a timely manner are deficient. These aspects have a negative impact on the ESP line: On the one hand, the strip flatness of the wedge zone cannot be predicted. This reduces the stability of the production process and affects the overall quality of the product. On the other hand, because the metal flow pattern during the FGC process is ambiguous, when producing two products with a large difference in thickness, the manufacturer performs a small FGC according to the plan for production stability. Thereby, many unplanned and unprofitable products are produced. This, in turn, reduces the economic efficiency and production speed.

Therefore, the study of the metal flow in the wedge zone of the FGC process is effective for obtaining better products and preventing production accidents. In addition, it provides a theoretical basis for developing the FGC transition system and improves the production efficiency. This study was aimed at proposing a mathematical model for predicting the flatness distribution of rolled specimens that accounts for the metal lateral flow in the wedge zone during the FGC process. The precision of the mathematical model was verified by finite element simulation. The effects of different rolling parameters on the metal flow in the wedge zone were investigated.

In this section, the theoretical background for calculating the flatness is introduced. The wedge zone flatness model with metal lateral flow is adopted in Sect. 2. A finite element model is established, and its accuracy is verified using the measured rolling force data in Sect. 3.1. The finite-element computations in Sect. 3.2 demonstrate the wedge zone flatness model. The effects of the reduction rate, roll crown, and friction coefficient on the flatness distribution of the wedge zone are discussed in Sect. 4. The main results and conclusions are presented in Sect. 5.

2 Establishment of the flatness calculation model

2.1 Description method of flatness defect

In the ESP endless-rolling process, the high-temperature strip appears to undergo plastic deformation under the action of a work roll. The strip outlet thickness profile is not a standard rectangle because of the non-uniform elastic deformation of the work roll and the edge drop effect in the rolled strip. Moreover, the thickness of each position in the width direction of the strip is different. Thus, the longitudinal fiber along the width direction of the strip is inconsistent, and a longitudinal residual stress may be produced. The critical

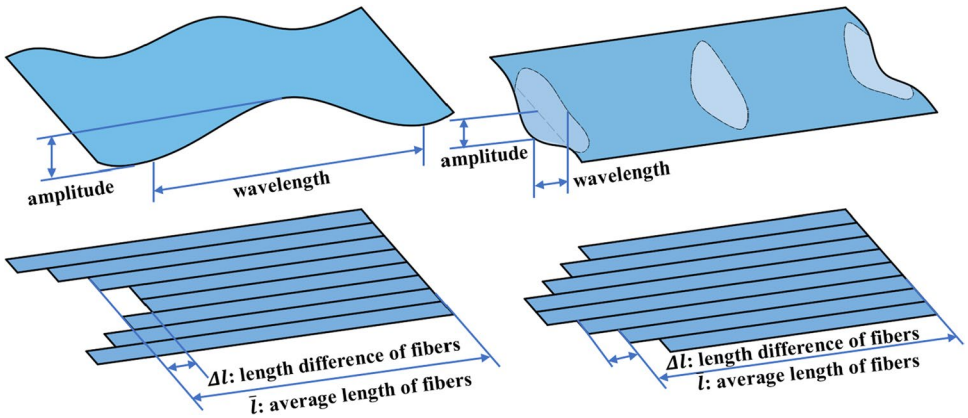


Fig. 3 Relationship between flatness defect and length difference of longitudinal fiber

buckling stress is exceeded when the residual stress and strip cause flatness defects during the rolling process [29, 30]. The flatness index I unit (IU) quantitatively assesses the flatness defects during the production process. According to the existing strip flatness distribution, the common defects of the strip shape include edge waviness, center waviness, M-mode waviness, and double-edge waviness [24]. As shown in Fig. 3, the flatness of the strip after rolling can be characterized as follows:

$$I_{\text{out}} = \frac{l - \bar{l}}{\bar{l}} \times 10^5 = \left(\frac{l}{\bar{l}} - 1\right) \times 10^5 \tag{1}$$

The average length of the longitudinal fibers is denoted as $\bar{l}$. Here, l is the longitudinal length at location y.

2.2 Establishment of flatness model

The coordinate system of the strip-wedge zone is illustrated in Fig. 4. The rolling direction of the strip is set as the longitudinal direction (x); the width direction is set as the transverse direction (y), and the thickness direction is set as the vertical direction (z). The strip-wedge zone is divided into a series of fiber strips of width dy during the FGC process. The length and thickness of the wedge zone are functions of y. A fiber element at a distance y from the center of the wedge zone was selected for the investigation. Its length before rolling is denoted as $L(y)$. The strip wedge zone head and tail thicknesses of the fiber element before rolling are denoted as $H_1(y)$ and $H_2(y)$, respectively. The corresponding quantities after rolling are denoted as $l(y)$, $h_1(y)$, and $h_2(y)$, respectively. $u_1(y)$ and $u_2(y)$ denote the metal transverse flows at the head and tail, respectively, of the strip-wedge zone at the stand exit. Therefore, the head width and tail of the wedge-zone fiber element are $dy + u_1(y + dy)$—$u_1(y)$ and $dy + u_2(y + dy)$—$u_2(y)$, respectively, after rolling.

In the design of varying thicknesses, the manufacturers drew from previous successful production experiences and rolled the transition zone of the FGC into a wedge zone to stabilize the production process. Therefore, in this study, the thickness along the rolling direction in the transition zone was assumed to be linearly distributed.

V_{in} and V_{out} denote the pre- and post-rolling volumes, respectively, of the fiber element. Then, the volumes V_{in} before rolling and V_{out} after rolling of the wedge zone fiber element are

$$V_{\text{in}} = \frac{H_1(y) + H_2(y)}{2} \bullet L(y) \bullet dy \tag{2}$$

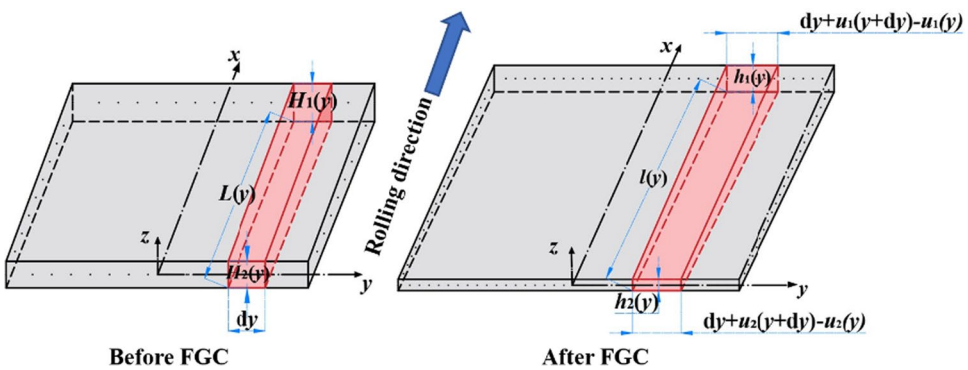


Fig. 4 3D coordinate system of rolled strip

$$V_{\text{out}} = \frac{\left\{\begin{array}{c} 2h_1(y) \bullet \left[u_1(y+dy) - u_1(y) + dy\right] + \\ 2h_2(y) \bullet \left[u_2(y+dy) - u_2(y) + dy\right] + \\ h_1(y) \bullet \left[u_2(y+dy) - u_2(y) + dy\right] + \\ h_2(y) \bullet \left[u_1(y+dy) - u_1(y) + dy\right] \end{array}\right\} \bullet l(y)}{6} \tag{3}$$

The following can be determined according to the volume conservation principle before and after rolling:

$$V_{\text{in}} = V_{\text{out}} \tag{4}$$

Combining Eqs. (2), (3), and (4), after rolling, it can be determined that the length of the wedge-zone fiber element is

$$l(y) = \frac{3\left[H_1(y) + H_2(y)\right] \bullet L(y) \bullet dy}{\left\{\begin{array}{l} h_1(y) \bullet \left[u_1(y+dy) - u_1(y) + dy\right] + h_2(y) \bullet \left[u_2(y+dy) - u_2(y) + dy\right] + \\ \left(h_1(y) + h_2(y)\right) \bullet \left[u_1(y+dy) - u_1(y) + dy + u_2 \bullet (y+dy) - u_2(y) + dy\right] \end{array}\right\}} \tag{5}$$

Equation (5) can be simplified as

$$l(y) = \frac{3\left[H_1(y) + H_2(y)\right] \bullet L(y)}{\left[2h_1(y) + h_2(y)\right] \bullet \left[1 + u_1'(y)\right] + \left[2h_2(y) + h_1(y)\right] \bullet \left[1 + u_2'(y)\right]} \tag{6}$$

We assume the average length of all the wedge-zone fiber elements after rolling to be $l(\bar{y})$ (i.e., the length of the wedge-zone fiber elements without flatness defects after rolling). Then, according to Eq. (5), the average length of the wedge-zone fiber element after rolling is

$$l(\bar{y}) = \frac{3\left[H_1(\bar{y}) + H_2(\bar{y})\right] \bullet L(\bar{y})}{\left[2h_1(\bar{y}) + h_2(\bar{y})\right] \bullet \left[1 + u_1'(\bar{y})\right] + \left[2h_2(\bar{y}) + h_1(\bar{y})\right] \bullet \left[1 + u_2'(\bar{y})\right]} \tag{7}$$

Combining Eqs. (6) and (7), the ratio of the average length of the wedge-zone fiber element to that of the wedge-zone fiber element without flatness defect is

$$\frac{l(y)}{l(\bar{y})} = \frac{\frac{\left[H_1(y)+H_2(y)\right] \bullet L(y)}{\left[H_1(\bar{y})+H_2(\bar{y})\right] \bullet L(\bar{y})}}{\frac{\left[2h_1(\bar{y})+h_2(\bar{y})\right] \bullet \left[1+u_1'(\bar{y})\right] + \left[2h_2(\bar{y})+h_1(\bar{y})\right] \bullet \left[1+u_2'(\bar{y})\right]}{\left[2h_1(y)+h_2(y)\right] \bullet \left[1+u_1'(y)\right] + \left[2h_2(y)+h_1(y)\right] \bullet \left[1+u_2'(y)\right]}} \tag{8}$$

If the rolling process is normal rolling with a constant thickness (i.e., $H_1(y) = H_2(y)$, $u_1(y) = u_2(y)$, and $h_1(y) = h_2(y)$), Eq. (8) can be simplified as

$$\frac{l(y)}{l(\bar{y})} = \frac{H(y) \bullet L(y)}{H(\bar{y}) \bullet L(\bar{y})} \frac{h(\bar{y}) \bullet \left[1 + u'(\bar{y})\right]}{h(\bar{y}) \bullet \left[1 + u'(\bar{y})\right]} \tag{9}$$

Equation (9) is consistent with the flatness calculation model that considers the metal lateral flow in a strip during the normal rolling process.

If the strip has no flatness defect and has good flatness before rolling, Eq. (8) can be simplified as

$$\frac{l(y)}{l(\bar{y})} = \frac{\left[2h_1(\bar{y}) + h_2(\bar{y})\right] \bullet \left[1 + u_1'(\bar{y})\right] + \left[2h_2(\bar{y}) + h_1(\bar{y})\right] \bullet \left[1 + u_2'(\bar{y})\right]}{\left[2h_1(y) + h_2(y)\right] \bullet \left[1 + u_1'(y)\right] + \left[2h_2(y) + h_1(y)\right] \bullet \left[1 + u_2'(y)\right]} \tag{10}$$

When the ratio of the width to the thickness of the wedge zone is significantly large during the FGC process (the width-to-thickness ratio is higher than 1000), the metal transverse flow in the wedge zone can be omitted. Subsequently, the ratio of the fiber strips after rolling can be simplified as

$$\frac{l(y)}{l(\bar{y})} = \frac{L(y)}{L(\bar{y})} \bullet \frac{H_1(y) + H_2(y)}{H_1(\bar{y}) + H_2(\bar{y})} \bullet \frac{h_1(\bar{y}) + h_2(\bar{y})}{h_1(y) + h_2(y)} \tag{11}$$

Combined with Eqs. (1) and (8), this yields the flatness distributions at different positions along the width direction of the wedge zone. A flatness calculation model of the wedge zone that considers the metal lateral flow of the strip can be obtained only by using the metal transverse flow curve and thickness distribution of the head and tail of the wedge zone before and after rolling.

During the FGC process, the fiber elements are mutually constrained. The strain of the strip after rolling in the wedge zone is

$$\varepsilon(y) = \frac{l(y) - l(\bar{y})}{l(\bar{y})} \tag{12}$$

where $\varepsilon(y)$ is the longitudinal strain of the wedge zone fiber element. Buckling deformation occurs when the residual stress in the strip exceeds the critical buckling stress. It is also known as a wave defect.

3 Model implementation and validation

Based on an ESP endless rolling line, a 3D thermomechanical coupling finite element model was established during the FGC process using ABAQUS finite element simulation software to obtain the metal transverse flow curve and flatness distribution of the wedge zone. In addition, the accuracy of the finite element and flatness calculation models was verified.

3.1 A 3D finite element model during the FGC process

A thermomechanical coupling 3D finite element model was established, as shown in Fig. 5. The following assumptions and simplifications were adopted for the established 3D finite element model to improve the efficiency of the numerical calculation:

(1) The strip and roll were entirely symmetric about the center of the wedge zone during the rolling process. Therefore, only a quarter model was established to replace the entire model.
(2) The roll was a rigid body rather than the original elastic–plastic body. The work roll profile controlled the elastic deformation of the roll.
(3) The strip was an ideal elastic–plastic body, and its material properties varied with temperature.
(4) To ensure the stability of the FGC process, a limited length (500 mm) was set up for the normal rolling process before the FGC starting point.

In the finite element model, the roll was coupled to a rigid body through a reference point, and the speed applied to the reference point caused the roll to rotate. The strip moved toward the roll at the initial speed and then was pulled into the roll gap by friction. The element type of the roll and strip was set to C3D8RT. The number of roll nodes was 11,000, and that of strip nodes was 312,417. The roll parameters, material properties, and main parameters of the strip-wedge zone are listed in Table 1. Figure 6 depicts the modeling procedure and validation accuracy.

The temperature is a key parameter that influences the material properties and metal flow during the FGC process. The influence of temperature variation on the strip material properties was considered in the finite element model to simulate the strip deformation in the FGC process more accurately. Q235B steel was selected for the rolling experiments in this study. The physical parameters of the low-carbon steel Q235B were acquired from Liu's research [31]. To determine the variation in the thermophysical parameters of Q235B steel with temperature, JMatPro software was used to simulate these parameters, and the calculated results were input into ABAQUS. The true stress–strain distribution curve of the strip steel with varying temperature is shown in Fig. 7.

Young's modulus, Poisson's ratio, thermal conductivity, and specific heat with respect to temperature are shown in detail in Fig. 8.

The influence of temperature on metal flow is highly important in the rolling process. The expansion coefficient of the metal varies with temperature. This results in different degrees of stress deformation at different positions on the strip. The temperature distribution of the strip along its width is shown in Fig. 9. For the F1 finishing mill stand, the temperature distribution in the width direction was approximately uniform (except at the edges) owing to the induction heating and dephosphorization experienced by the billet. A small number of temperature measurement points were considered during the actual production. A smooth temperature distribution curve was obtained in the strip width direction through data fitting, as shown in Fig. 9. The convective heat transfer coefficient between the strip and surrounding environment was 0.17 kW/ (m^2·℃), the surface thermal radiation coefficient of the strip was 0.8, and the heat exchange coefficient between the roll and strip was 30 kW/(m^2·℃). Considering the heat generated by the frictional force and deformation work, the conversion coefficient was set to 0.9, the ambient temperature around the strip was set to 25 ℃, and the roll temperature was set to 150 ℃.

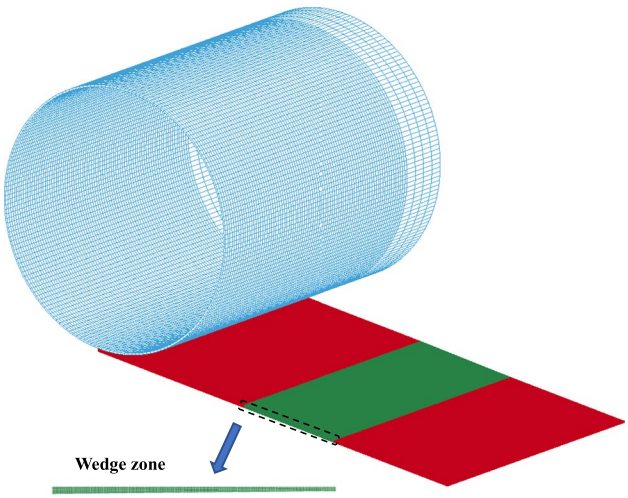


Fig. 5 3D finite element model of work roll and strip

Table 1 Modeling simulation parameters for rolling

Parameter	Value	Unit
Work roll length	1950	Mm
Work roll diameter	750	Mm
Work roll elasticity modulus	210	GPa
Work roll density	7850	kg/m^3
Wedge zone entry head thickness	17	mm
Wedge zone entry tail thickness	10	mm
Wedge zone entry length	311.3	mm

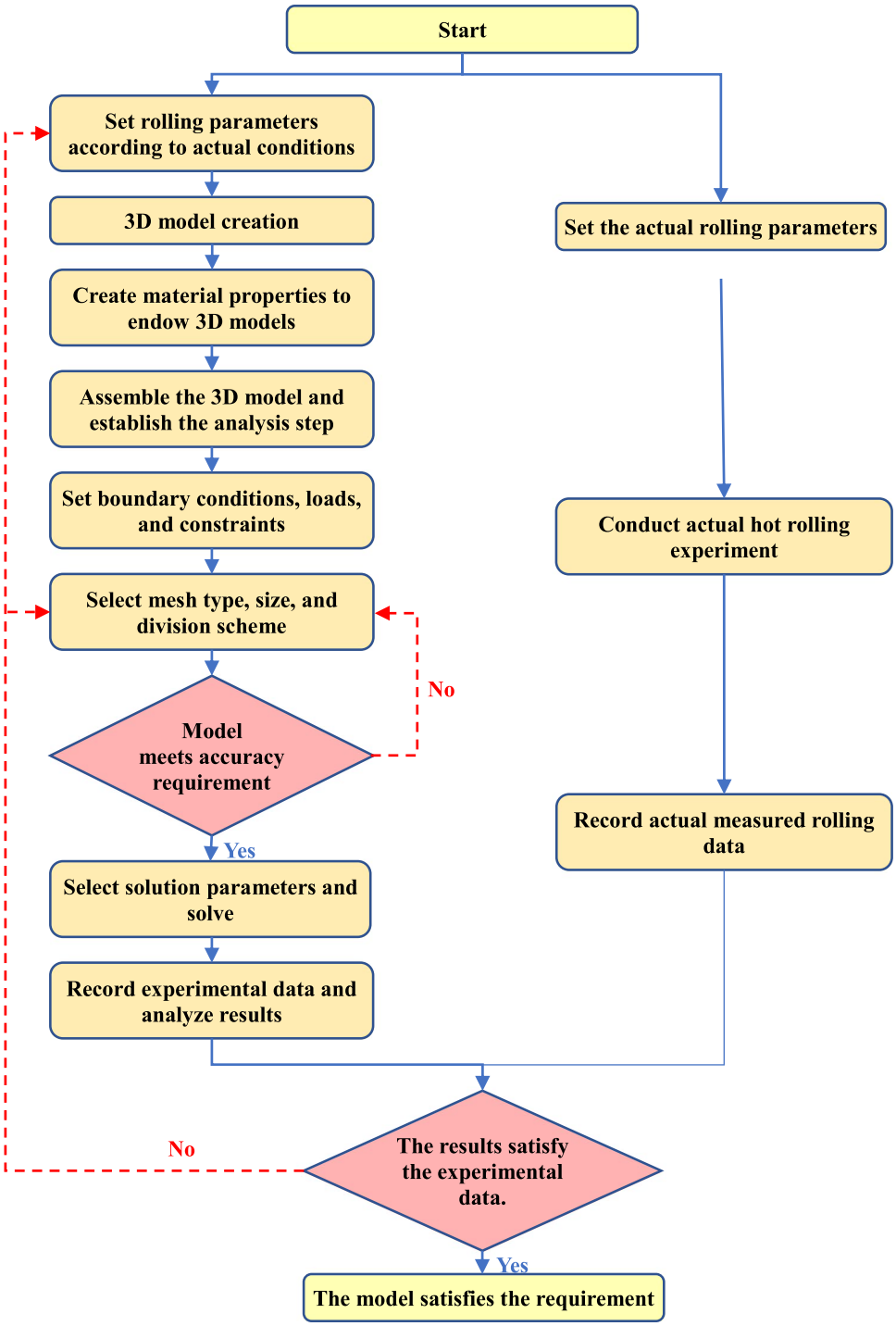


Fig. 6 Flow chart of simulation modeling and analysis for strip rolling

3.2 Validation of finite element model and wedge zone flatness model

To verify the accuracy of the finite element model, the rolling force–time history curve of the finite element simulation results was extracted, and the simulated rolling force was compared with the average value of the measured rolling force. The comparison results are presented in Table 2. A finite element simulation of the rolling force–time history curve is shown in Fig. 10. As shown in Fig. 10, as the strip is bitten into by the work roll, the rolling force starts to increase, and when the rolling process stabilizes,

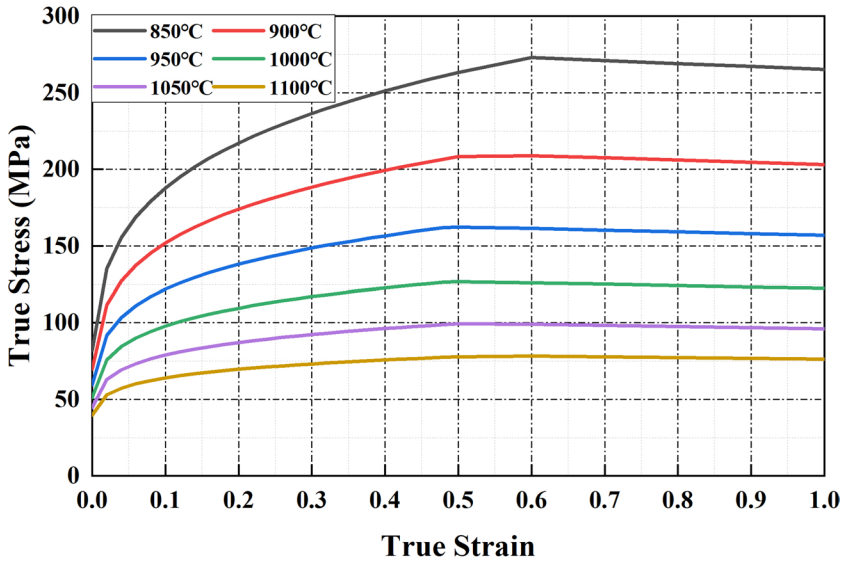


Fig. 7 True stress–strain curve of strip

the rolling force also starts to become stable. The rolling force began to increase when the strip arrived at the FGC starting point. Finally, the rolling force stabilized at another value when the FGC stopping point was reached. The finite element values agree well with the measured values. Thus, the calculation results of the simulation model comply completely with the accuracy standards of the current study.

A finite-element model verified earlier for accuracy through simulation experiments was utilized to evaluate the accuracy of the mathematical model. The metal transverse flow curves and thickness distributions at the head and tail of the wedge zone were extracted from the model. The metal transverse flow curves from the two areas were fitted and introduced into the mathematical model. The metal distribution in the wedge zone was calculated and

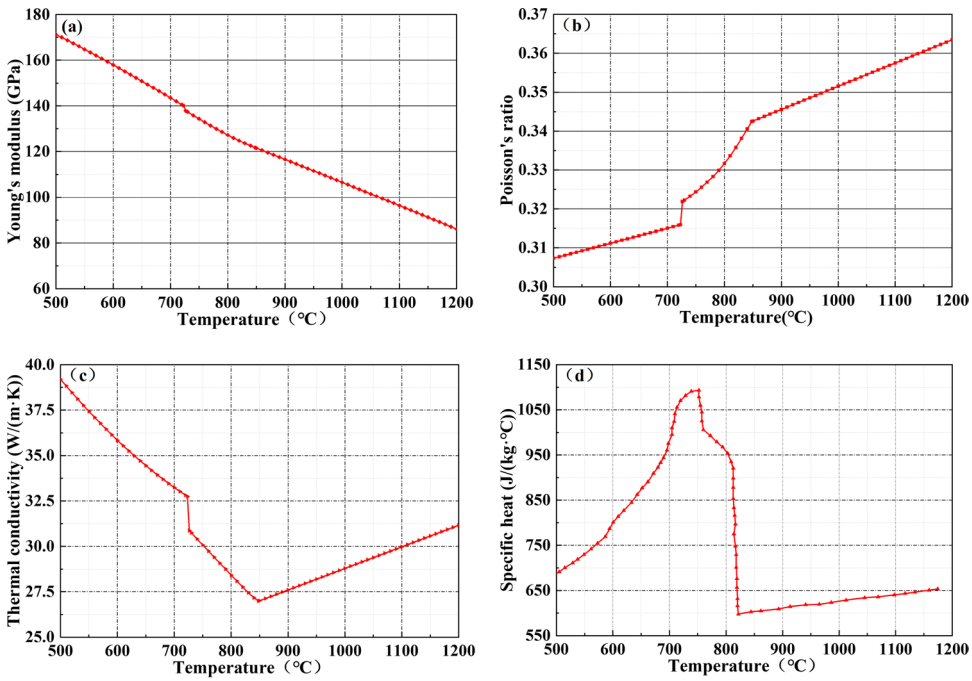


Fig. 8 Physical parameters of strip: **a** Young's modulus; **b** Poisson's ratio; **c** thermal conductivity; **d** specific heat

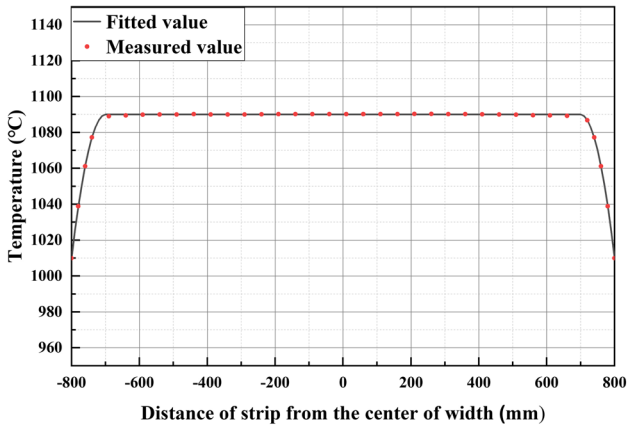


Fig. 9 Initial temperature distribution of strip

compared with the measured flatness value in the simulation, as shown in Fig. 11.

From the above figure, it can be observed that along the width direction of the strip, the values calculated using the prediction model significantly match the actual measured strip flatness curve. In the width direction, the prediction model exhibited an exceptionally high accuracy at the center of the strip. However, the deviation of the calculated results of the prediction model from the actual measured values increased gradually as the observation point shifted from the center of the strip to the edge. However, it still maintained a high level of accuracy. This was because the amount of roll reduction in the FGC process varied, and the metal flow at the edge of the strip varied more significantly than that at the center of the strip. In particular, the metal flow in the thickness direction of the strip also showed differences. This resulted in a more complex distribution of metal at the edge and, therefore, a marginal deviation in the calculated results.

Table 2 Comparison between measured and FEM values of rolling force

Entry thickness (mm)		Exit thickness (mm)		Before FGC			After FGC		
				Rolling force (kN)		Error (%)	Rolling force (kN)		Error (%)
Head	Tail	Head	Tail	Measured values	FEM values		Measured values	FEM values	
17	10	8.5	4	26,307	26,724	1.59	31,098	31,823	2.36

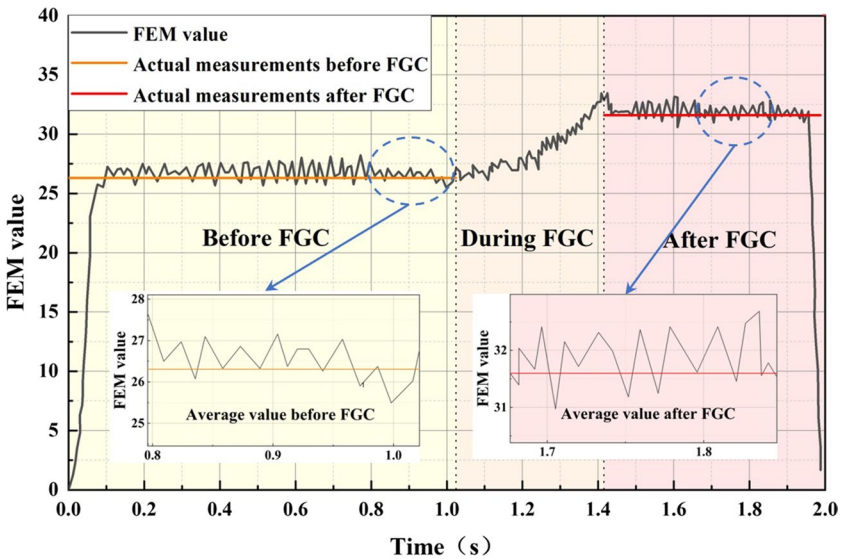


Fig. 10 Time history of rolling force curve during FGC

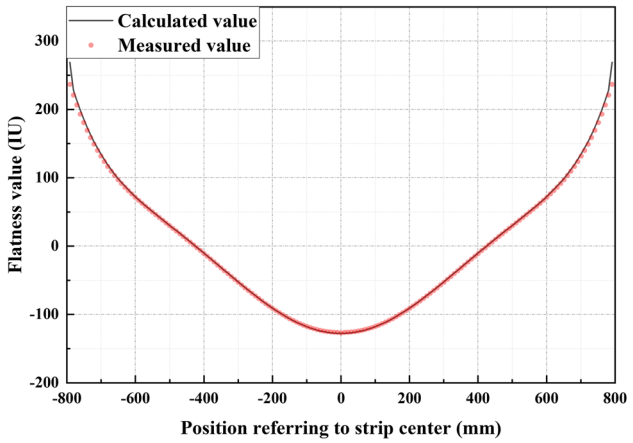


Fig. 11 Comparisons of flatness distribution between measured value and calculated value

4 Effects of different parameters on lateral spread and flatness

The effects of different head and tail reduction rates in the wedge zones on the metal transverse flow and flatness distribution were analyzed to further study the FGC process. During the rolling process, the roll crown varied gradually owing to the wear and thermal expansion of the roll. Therefore, the influence of different roll crowns on the metal transverse flow and flatness distribution of the wedge zone was studied. Simultaneously, the influence of different friction coefficients between roll and wedge zones on the metal transverse flow law and flatness distribution was analyzed.

4.1 Influence of reduction rate on lateral spread and flatness

Different transitional systems have different reduction rates. This results in different metal transverse flows and flatness distributions of the wedge zone during the FGC process. Therefore, a simulation analysis of different head and tail reduction rates in the wedge zone was performed. The wedge zone head and tail thicknesses before rolling were 17 mm and 10 mm, respectively. The length of the wedge zone was 311.3 mm. A controlled variable experiment was conducted on the model. Here, the reduction rate of the head and tail of the wedge zone increased from 30 to 50% when the roll crown was 150 μm. The lateral spread curve of the wedge zone after rolling was obtained, as shown in Fig. 12.

As shown in Fig. 12a, the metal transverse flow from the middle to the edge increased gradually in the wedge zone along the width direction during the FGC process. This was because the transverse resistance was higher than the longitudinal resistance in the middle metal of the wedge zone. Therefore, most of the metal flowed in the rolling direction. In addition, the closer to the wedge zone edge, the lower was the lateral resistance. The metal lateral flow increased gradually. This also caused the lateral flow slope of the edge metal to be more prominent than that of the middle metal.

Combined with Fig. 12a, b, and c, the metal lateral flow increased with an increase in the reduction rate in the wedge zone. The maximum metal lateral displacements at the head and tail of the wedge zone were 0.89 mm and 1.00 mm, respectively, when the reduction rate was 30%, and were 1.34 mm and 1.44 mm, respectively, when the reduction rate was 50%. The reduced volume of the wedge zone increased as the reduction rate increased. According to the law of volume invariance before and after rolling, the lateral flow of metal increased. Moreover, the slope of the metal transverse flow at the wedge zone edge increased gradually as the reduction rate increased. This also resulted in a gradual increase in the metal transverse flow. In addition, combined with Fig. 12, it can be observed that the easy transverse flow zone in the middle of the wedge zone decreased gradually with an increase in the reduction rate. This was because of the increase in the reduction rate. The rolling deformation zone became larger. This resulted in a higher resistance to the metal particle flow in the wedge zone. Therefore, the easy flow zone in the middle of the wedge zone became smaller gradually.

To obtain the flatness distribution in the wedge zone, the metal transverse flow curves at the head and tail for different reduction rates were extracted and were fitted by the quintic polynomial (see Eq. (13)). The fitting results for different reduction rates are shown in Fig. 13. Here, $R^2 > 0.995$. This indicates that the metal transverse flow curves at the head and tail of the wedge zone are well fitted by the quintic polynomial. This satisfies the accuracy required by calculation. An analysis of these figures also verifies the conclusion that the easy flow area gradually became smaller with an increase in the reduction rate.

$$\begin{cases} u(y) = A_0 + A_1 \bullet y + A_2 \bullet y^2 + A_3 \bullet y^3 + A_4 \bullet y^4 + A_5 \bullet y^5 \\ u'(y) = A_1 + 2A_2 \bullet y + 3A_3 \bullet y^2 + 4A_4 \bullet y^3 + 5A_5 \bullet y^4 \end{cases} \tag{13}$$

where A_0–A_5 are the fitting coefficients of the fitted curves. The fitting coefficients of the fifth-order polynomial fitting were obtained by extracting the metal transverse flow curves

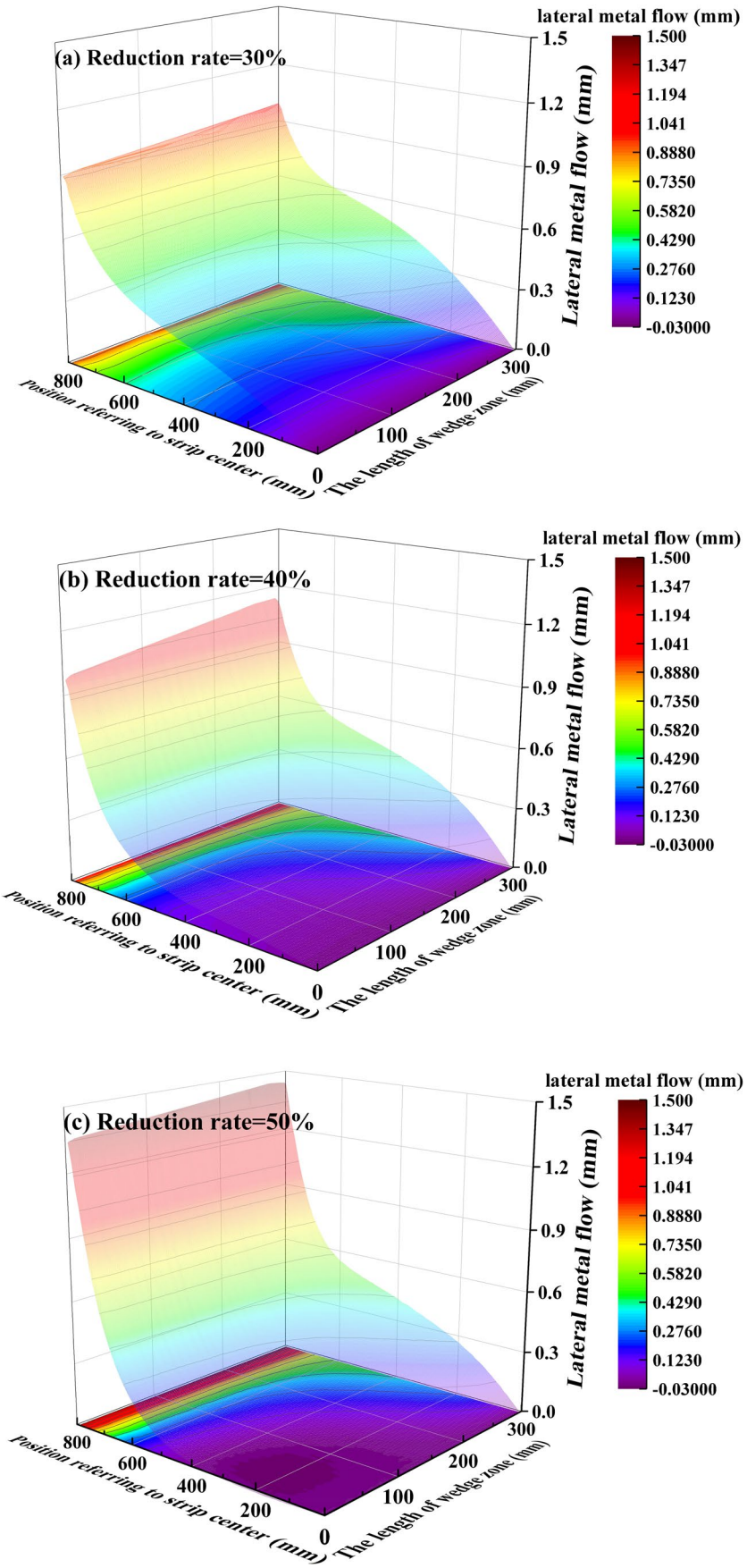


Fig. 12 Effect of reduction rate of strip on metal lateral flow: **a** reduction rate = 30%; **b** reduction rate = 40%; **c** reduction rate = 50%

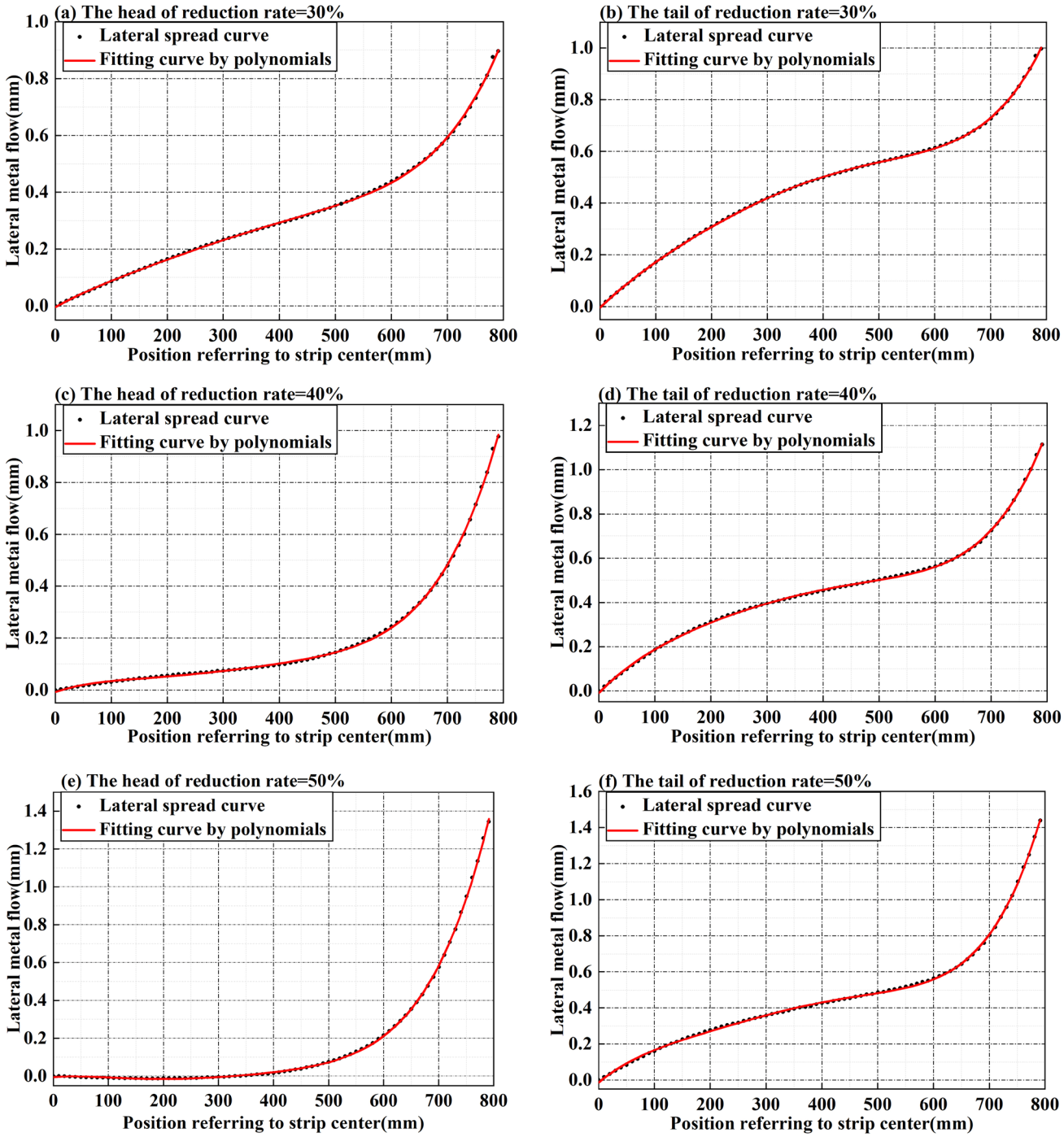


Fig. 13 Lateral spread curves for different reduction rates: **a** the head of reduction rate = 30%; **b** the tail of reduction rate = 30%; **c** the head of reduction rate = 40%; **d** the tail of reduction rate = 40%; **e** the head of reduction rate = 50%; **f** the tail of reduction rate = 50%

of the head and tail portions of the metal at different reduction rates (see Table 3).

The derivatives of the metal transverse flow curves at the head and tail of the wedge zone were input into the flatness calculation model of the wedge zone introduced into the metal transverse flow in the FGC process. The flatness distributions of the wedge zones at different reduction rates are shown in Fig. 14. According to this figure, the flatness value at the edge of the wedge zone was positive and that at the center was negative when the roll crown was − 150 μm.

Table 3 Fitting coefficients of fitted curves under different reduction rates

Wedge zone	Reduction rate	$A_0\times10^{-3}$	$A_1\times10^{-3}$	$A_2\times10^{-6}$	$A_3\times10^{-8}$	$A_4\times10^{-11}$	$A_5\times10^{-14}$	R_s^2
Head	30%	−4.47	1.1	−2.55292	9.03765E^{-1}	−1.73544	1.27583	0.9997
Tail	30%	−4.62	2.09	−4.32271	1.23992	−2.35516	1.65894	0.9999
Head	40%	−8.06	7.55046E^{-1}	−4.89745	1.88195	−3.31333	2.29792	0.99958
Tail	40%	−9.94	2.63	−8.68083	2.40207	−3.87231	2.50124	0.99975
Head	50%	−6.01	2.545E^{-1}	−4.48063	2.22748	−4.34732	3.2153	0.9998
Tail	50%	−1.1428E^{1}	2.64	−1.17714E^{1}	4.06293	−7.00164	4.53082	0.99966

The wedge zone exhibited edge waviness. This was because the metal at the edge had a more significant elongation than that in the middle part of the wedge zone. In addition, the flatness value increased gradually as the reduction rate of the wedge zone decreased. The maximum flatness of the wedge zone was 533 IU and 304 IU at the reduction rates of 30% and 50%, respectively. This was because with a decrease in the reduction rate, the easy transverse flow zone became larger, and the metal transverse flow in the middle of the wedge zone became more intense, which caused a gradual decrease in the elongation of the metal in the middle. Thus, the flatness value in the wedge zone increased gradually.

4.2 Influence of roll crown on flatness and lateral spread

The crown of the roll varies gradually owing to factors such as the roll wear, non-uniform thermal expansion, and stress deformation. It is an important factor affecting the shape distribution during the rolling process. It is necessary to study and discuss the distribution law of the metal lateral flow and flatness in the wedge zone of the strip steel during FGC under different rolling crowns. Therefore, a finite element simulation model of different roll crowns was established.

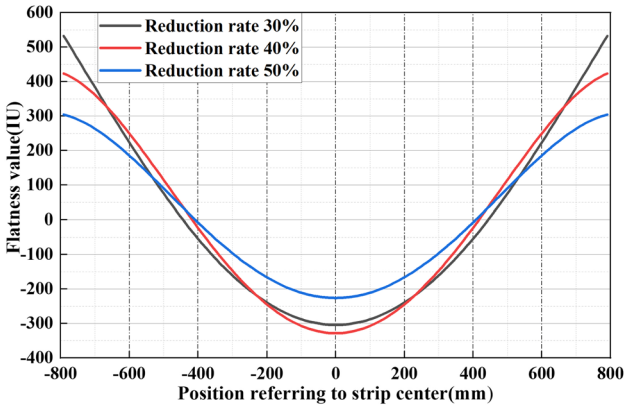


Fig. 14 Flatness distribution for different reduction rates

Similarly, the metal lateral flow curve with the roll crown varying from 0 to − 150 μm was extracted (see Fig. 15).

Combined with Fig. 15, it can be observed that with a decrease in the roll crown, the metal transverse flow in the wedge zone decreased gradually, and the easy flow zone of the metal transverse flow became smaller. The decreased rate of the metal transverse flow at the head was less than that at the tail in the wedge zone. The maximum metal lateral displacements at the head and tail of the wedge zone were 1.34 mm and 1.44 mm, respectively, when the roll crown was − 150 μm, and were 0.78 mm and 0.03 mm, respectively, when the roll crown was 0 μm. This was because the reduction ratio of the edge relative to the middle decreased gradually with a decrease in the roll crown. This weakened the capability of the metal lateral flow in the edge, and the metal lateral flow reduced gradually. In addition, the exit thickness of the head was larger than that of the tail in the wedge zone. When the crown of the roll varied, the variation rate of the crown ratio at the head was less than that at the tail in the wedge zone. This caused the reduction rate of the metal transverse flow at the head to be less than that at the tail in the wedge zone.

The metal lateral flow curves for the head and tail of the wedge zone under different roll crowns were extracted, and the strip flatness distributions in the wedge zone under different roll crowns were calculated. The strip flatness distribution curve for the wedge zone is shown in Fig. 16. With the decrease in roll crown, the flatness value varied from positive to negative at the edge in the wedge zone, and the flatness value varied from negative to positive at the middle in the wedge zone. When the roll crown was − 150 μm, the maximum flatness value at the edge of the wedge zone was 470 IU, and the minimum flatness value at the middle of the wedge zone was − 534 IU. The wedge zone exhibited edge waviness. When the roll crown was − 50 μm, the maximum flatness value at the edge of the wedge zone was 58 IU, the minimum flatness value at the quarter region in the wedge zone was − 67 IU, and the flatness value at the middle of the wedge zone was 32 IU. The wedge zone exhibited mid-edge

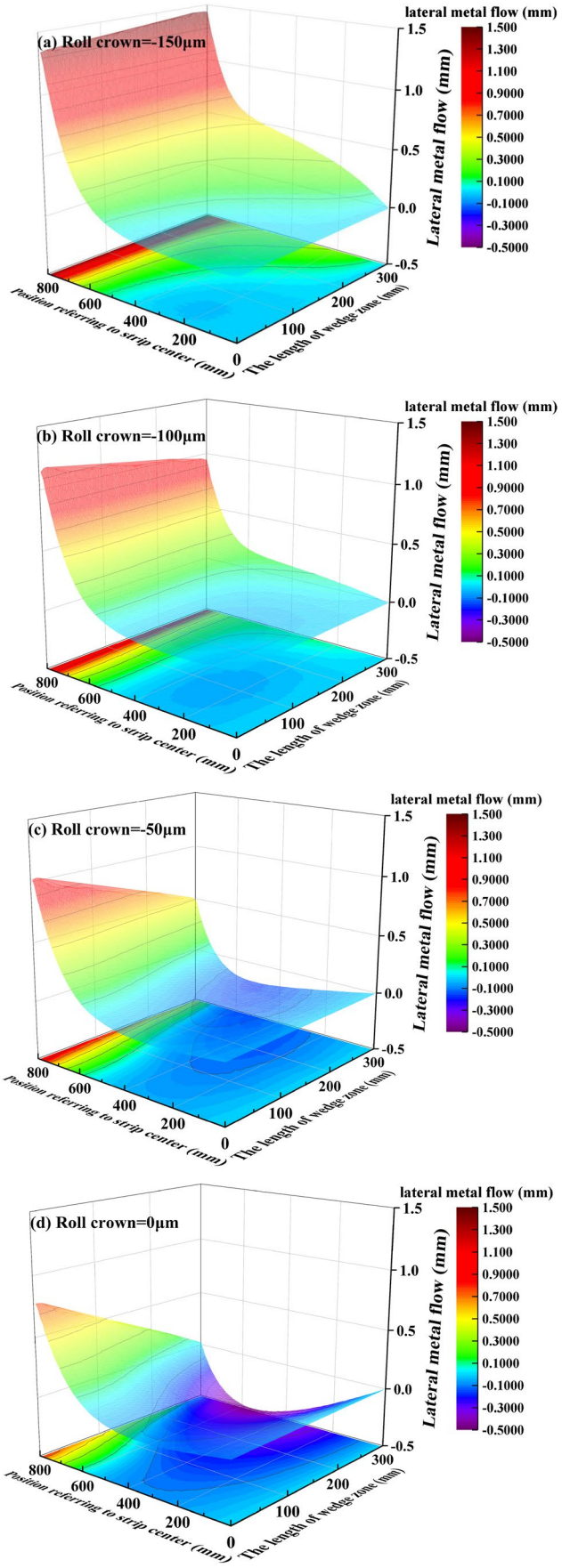


Fig. 15 Lateral spread curves for different roll crowns: **a** roll crown = -150μm; **b** roll crown = -100μm; **c** roll crown = -50μm; **d** roll crown = 0μm

compound waviness. When the roll crown was 0 μm, the maximum flatness value at the middle of the wedge zone was 227 IU, and the minimum flatness value at the edge in the wedge zone was – 257 IU. The wedge zone exhibited center-waviness behavior.

This was because the reduction rate of the edge relative to the middle decreased gradually as the roll crown decreased. This resulted in a gradual decrease in the elongation of the edge relative to the middle. The central elongation of the wedge zone exceeded that of the edge when the roll crown was reduced to a particular value. Therefore, the flatness distribution transitioned gradually from edge waviness to center waviness with a decrease in the roll crown.

4.3 Influence of friction coefficient on lateral spread and flatness

In the FGC process, the friction force between the strip and roll also differs owing to the plastic deformation of the strip, wear of the roll surface, and differences in the lubricating fluid used between the roll and strip. To study the effects of different friction coefficients on the metal transverse flow and flatness distribution in the wedge zone during FGC, three rolling schedules with friction coefficients of 0.2, 0.3, and 0.4 were selected for the simulation, and the contour graph of the metal transverse flow was obtained (see Fig. 17).

As shown in Fig. 17, as the friction coefficient increased, the metal transverse flow decreased gradually, and the easy flow zone gradually became larger in the strip wedge zone. The maximum metal lateral displacements at the head and tail of the wedge zone were 0.96 mm and 0.13 mm, respectively, when the friction coefficient was 0.2, and were 0.56 mm and – 0.14 mm, respectively, when the friction coefficient was 0.4. When the friction coefficient increased, the increase in the resistance of the metal lateral flow became larger than that for the metal longitudinal flow in the wedge zone. Therefore, the metal lateral flow decreased gradually in the wedge zone as the friction coefficient increased.

As shown in Fig. 18, when the roll crown was 0 μm, the flatness value in the middle of the strip was positive and decreased with an increase in the friction coefficient. Meanwhile, the flatness value of the edge increased with an increase in the friction coefficient and was negative. The maximum flatness value in the wedge zone was 247 IU when the friction coefficient was 0.2 and decreased to 176 IU when it was 0.4. This demonstrated that the likelihood of a middle wave defect in the wedge zone of the strip reduced

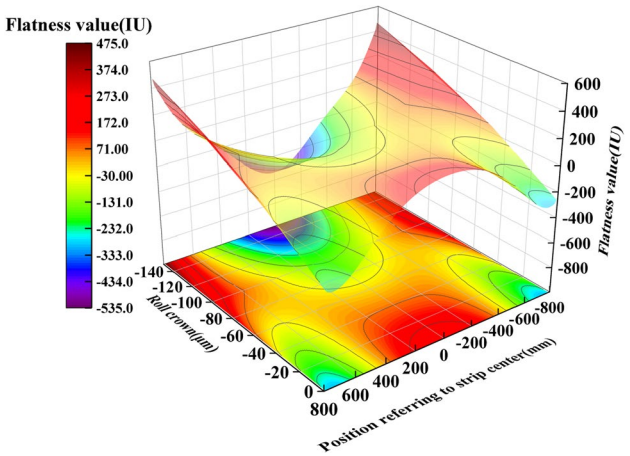


Fig. 16 Flatness distribution for different roll crowns

with the increase in the friction coefficient. As the friction coefficient increased, the easy-flow area decreased in the middle of the wedge zone, and the relative metal lateral flow in the middle and edge of the wedge zone decreased. Therefore, the flatness value decreased in the middle of the wedge zone, whereby the trend of the occurrence of central waviness becomes more inconspicuous.

5 Conclusions

In this study, a mathematical model was developed to calculate the flatness of the wedge zone while considering the metal transverse flow. The metal transverse flow and flatness distribution in the wedge zone during the FGC were studied. The conclusions of this analysis are as follows:

(1) A mathematical model for calculating the flatness of the wedge zone considering the metal transverse flow was established based on the volume conservation principle before and after rolling. This provided a method for calculating flatness and studying the influence of the metal transverse flow on the flatness distribution of the wedge zone during the FGC process.

(2) A 3D finite element model of the FGC process was established by comparing the FEM simulation results with the actual production rolling force results. The average rolling force deviation of FGC before and after FGC is 1.59% and 2.36%, respectively. A comparison of the calculated values of the model with the measured flatness values revealed the two to be in good agreement. This demonstrated that the recommended mathematical model for the flatness of the wedge zone was accurate.

(3) According to the analysis of the impact of the reduction rate on the flatness of the metal and the lateral flow, an increase in the reduction rate caused an increase in the transverse displacement of the metal, and the flatness of the strip wedge zone decreased from 533 to 304 IU. Meanwhile, the flatness distribution transitioned gradually from edge waviness to center waviness as the roll crown decreased. When the friction coefficient decreased, the metal lateral flow increased gradually, and the flatness of the wedge zone increased from 176 to 247 IU.

Author contribution All authors contributed to the study conception and design. Material preparation, data collection, and analysis were performed by Wen Peng, Qinglong Wang, and Jie Sun. The first draft of the manuscript was written by Xi Chen and Zilong Wan; Dianhua Zhang provides important guidance on some of the issues that arise, and all authors commented on previous versions of the manuscript. All authors read and approved the final manuscript.

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Declarations

Competing interests The authors declare no competing interests.

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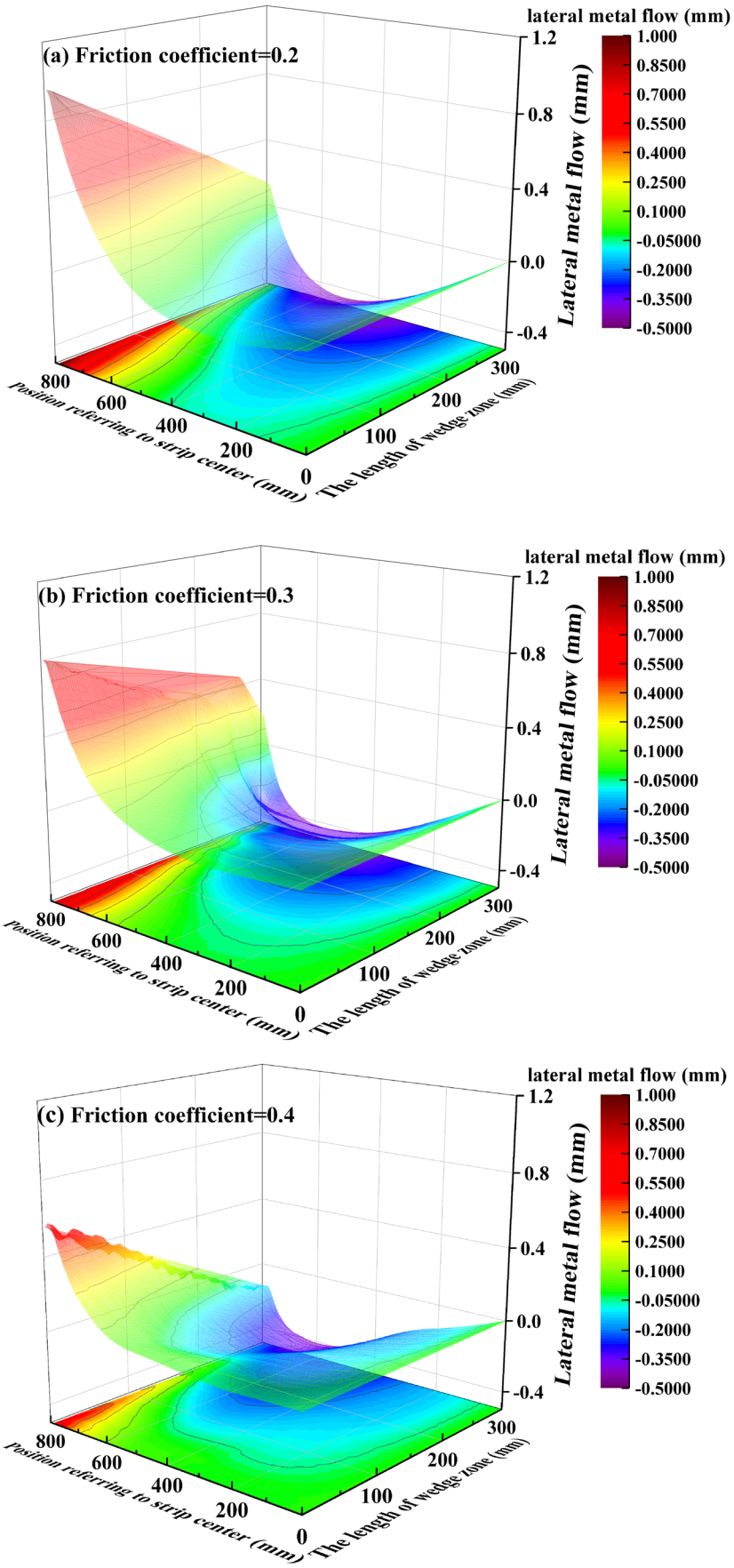


Fig. 17 Lateral spread curves for different friction coefficients: **a** friction coefficient = 0.2; **b** friction coefficient = 0.3; **c** friction coefficient = 0.4

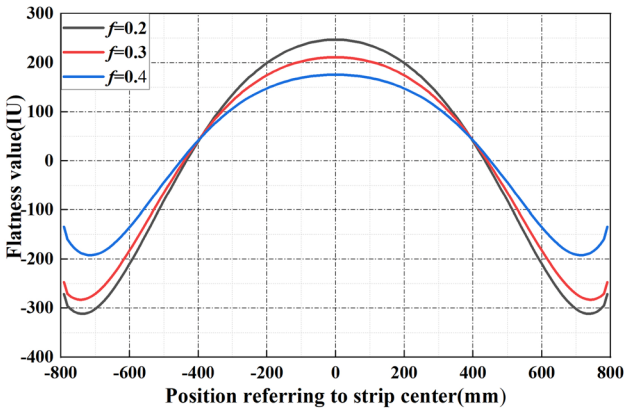


Fig. 18 Flatness distribution of different friction coefficient

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